

Kieran A. Murphy

Assistant Professor, Computer Science
Affiliated Faculty, Department of Data Science
New Jersey Institute of Technology

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Research: Develops information-theoretic methods for understanding and controlling information flow in machine learning systems, with applications to interpretability and scientific discovery.

EDUCATION

University of Chicago, Chicago IL

Ph.D., Physics, June 2019

M.S., Physical Sciences, June 2014

University of California, Berkeley, Berkeley CA

B.A. with High Distinction, Physics (Computer Science emphasis), May 2013

RESEARCH POSITIONS

Assistant Professor

Sep 2025 - Present

New Jersey Institute of Technology, Dept. of Computer Science, Newark NJ

Postdoctoral Fellow

July 2021 - Aug 2025

University of Pennsylvania, Department of Bioengineering, Philadelphia PA

Advisor: Dani S. Bassett

Developed machine learning methods grounded in information theory to analyze complex systems.

AI Resident

Oct 2019 - June 2021

Google Research, New York NY

Mentors: Ameesh Makadia, Srikumar Ramalingam

Worked on weakly-supervised representation learning and 3D perception.

Research Assistant

Oct 2013 - June 2019

University of Chicago, Department of Physics, Chicago IL

Advisor: Heinrich M. Jaeger

Investigated disordered systems and granular materials using experimental and computational methods.

Undergraduate Research Assistant

June 2012 - June 2013

Lawrence Berkeley National Laboratory, Berkeley CA

Undergraduate Research Assistant

May 2011 - Aug 2011

University of California, Davis, Davis CA

PUBLICATIONS (Google Scholar)

Journal Articles

14. **Murphy, K.A.**, Zhang, Y., Bassett, D.S., “Surveying the space of descriptions of a composite system with machine learning.” *Phys. Rev. Lett.* (2025). [\[link\]](#)
Editors’ Suggestion.
13. Ouellet, M., Bassett, D.S.*, Bassett, L.C.*, **Murphy, K.A.***, Patankar, S.P.*, “Mechanical prions as self-assembling microstructures.” *Newton* 1, 100098 (2025). [\[link\]](#)

12. **Murphy, K.A.**, Dillavou, S., Bassett, D.S., “Comparing the information content of probabilistic representation spaces.” *Transactions on Machine Learning Research (TMLR)* 2025) [\[link\]](#) [\[code\]](#)
11. **Murphy, K.A.**, Bassett, D.S., “Machine-learning optimized measurements of chaotic dynamical systems via the information bottleneck.” *Phys. Rev. Lett.* 132, 197201 (2024) [\[link\]](#) **Editors’ Suggestion, PRL Collection of the Year 2024.**
10. **Murphy, K.A.**, Bassett, D.S., “Information decomposition in complex systems via machine learning.” *Proc. Natl. Acad. Sci.* 121 (2024). [\[link\]](#) [\[project page\]](#)
9. Reyes-Martinez, M.A., Chan, E., Soles, C.L., Han, E., **Murphy, K.A.**, Jaeger, H.M., Reid, D.R., de Pablo, J.J., “Tuning the mechanical impedance of disordered networks for impact mitigation.” *Soft Matter* 18, 2039-2045 (2022). [\[link\]](#)
8. **Murphy, K.A.**, Kruppe, J., Jaeger, H.M., “Memory in nonmonotonic stress relaxation of a granular system.” *Phys. Rev. Lett.* 124, 168002 (2020). [\[link\]](#) **Editors’ Suggestion.**
7. **Murphy, K.A.**, “Granular materials as a model for how disordered systems adapt” Dissertation, University of Chicago, 2019. [\[link\]](#)
6. **Murphy, K.A.**, MacKeith, A., Roth, L.K. Jaeger, H.M., “The intertwined roles of particle shape and surface roughness in controlling the shear strength of a granular material.” *Granular Matter* 21, 72 (2019). [\[link\]](#)
5. Lim, M.X., **Murphy, K.A.**, Jaeger, H.M., “Edges control clustering in levitated granular matter.” *Granular Matter* 21, 77 (2019). [\[link\]](#)
4. **Murphy, K.A.**, Dahmen, K.A., Jaeger, H.M., “Transforming mesoscale granular plasticity through particle shape.” *Phys. Rev. X* 9, 011014 (2019). [\[link\]](#)
3. **Murphy, K.A.**, Roth, L.K., Peterman, D., Jaeger, H.M., “Aleatory architectures based on jamming.” *Architectural Design* 187, 74-81 (2017). [\[link\]](#) [\[pdf\]](#)
2. **Murphy, K.A.**, Roth, L.K., Jaeger, H.M., “Adaptive granular matter,” in *Active Matter*, ed. Skylar Tibbits, MIT Press, Cambridge (2017).
1. **Murphy, K.A.**, Reiser, N., Choksy, D., Singer, C.E., Jaeger, H.M., “Freestanding load bearing structures with Z-shaped particles.” *Granular Matter* 18, 26:21-29 (2016). [\[link\]](#)

AI/ML Conference Proceedings (peer-reviewed)

4. Patankar, S., Ouellet, M., Cervino, J., Ribeiro, A., **Murphy, K.A.**, Bassett, D.S., “Intrinsically motivated graph exploration using network theories of human curiosity.” (*Learning on Graphs Conference* 2023) [\[OpenReview link\]](#) [\[arxiv link\]](#)
3. **Murphy, K.A.**, Bassett, D.S., “Interpretability with full complexity by constraining feature information.” (*International Conf. on Learning Representations (ICLR)* 2023) [\[OpenReview link\]](#) [\[arxiv link\]](#) [\[project page\]](#)
2. **Murphy, K.A.**, Jampani, V., Ramalingam, S., Makadia, A., “Learning ABCs: Approximate Bijective Correspondence for isolating factors of variation.” (*Conf. on Computer Vision and Pattern Recognition (CVPR)* 2022) [\[link\]](#) [\[project page\]](#) **Selected for oral presentation.**
1. **Murphy, K.A.***, Esteves, C.*, Jampani, V., Ramalingam, S., Makadia, A., “Implicit representation of probability distributions on the rotation manifold.” (*International Conf. on Machine Learning (ICML)* 2021). [\[link\]](#) [\[project page\]](#)

Workshop Papers (light peer-review)

4. **Murphy, K.A.**, “InfoChess: A Game of Adversarial Inference and a Laboratory for Quantifiable Information Control.” “Adaptive and Learning Agents Workshop” AAMAS 2026 workshop. [\[link\]](#)
3. **Murphy, K.A.**, Yin, Z., Bassett, D.S., “Which bits went where? Past and future transfer entropy decomposition with the information bottleneck.” “Machine learning and the physical sciences” NeurIPS 2024 workshop. [\[link\]](#)
2. **Murphy, K.A.**, Bassett, D.S., “Where is the information in data?” IEEE VIS 2024 workshop, “Visualization for AI Explainability” [\[link to interactive tutorial\]](#)
1. **Murphy, K.A.**, Bassett, D.S., “Characterizing information loss in a chaotic double pendulum with the Information Bottleneck.” NeurIPS 2022 workshop, “Machine learning and the physical sciences.” *Selected for oral presentation.* [\[link\]](#)

Preprints and Papers Under Review

1. **Murphy, K.A.**, “From independent patches to coordinated attention: Controlling information flow in vision transformers.” arXiv 2602.04784 (Under review) [\[link\]](#)

TEACHING

New Jersey Institute of Technology, Department of Computer Science Instructor of Record

Explainable AI (CS 485/698, Fall 2025)

[\[Syllabus\]](#)

Designed and taught a new undergraduate/graduate course on explainable machine learning, with original interactive demos and projects emphasizing interpretability, transparency, and performance trade-offs.

Explainable AI (CS 785, Spring 2026)

[\[Syllabus\]](#)

FELLOWSHIPS AND AWARDS

- *AI x Science Postdoctoral Fellow*, University of Pennsylvania (2024-2025)
- *Data Science Postdoctoral Fellow*, Data Driven Discovery Initiative, University of Pennsylvania (2022-2024)
- *Best Poster Award*, Gordon Research Conference on Granular Matter (2018).

PRESS

- NJIT News, “Computer Science and Data Science Tame AI at NJIT” [\[link\]](#) (March 26, 2026).
- Penn Today, “AI x Science Postdoctoral Fellows collaborate across disciplines” [\[link\]](#) (June 9, 2025).
- Penn Today, “Unlocking the mechanics of protein misfolding” [\[link\]](#) (June 2, 2025).
- Penn Engineering Today, “Measuring Chaos: Using Machine Learning to Satisfy Our Need to Know” [\[link\]](#) (July 22, 2024).
- *Quantum Photonics* on Clubhouse (audio app) “Can deep learning break down what it learns?” (May 13, 2022).

INVITED
PRESENTATIONS

- The New Jersey workshop on Machine Learning and Information Theory, “Making information flow in neural networks explicit and controllable” (May 1, 2026).
- Partnership for Research and Education in Materials, San Juan, Puerto Rico, “Listening to deep learning models talk about data” (May 16, 2025).
- ‘Information theory, machine learning, and network sciences (infoLeNS)’ seminar series, University of Pennsylvania, “Searching through distributed compression schemes to answer ‘where is the information in data?’ ” (April 12, 2024).
- Penn ASSET/IBI Symposium on Trustworthy AI for Health Care, University of Pennsylvania, “Optimizing clinical monitoring for delivery room resuscitation using novel interpretable AI” (Oct 6, 2023, presented with Elizabeth Foglia, Perelman School of Medicine).
- James Franck Institute Special Seminar, University of Chicago, “All information is not created equal: Using machine learning to approximate relationships for interpretability and information theoretic analyses of complex systems.” (Feb 9, 2023).

SERVICE

Academic service

- **Reviewer** – NeurIPS, ICML, CVPR, ICLR, TMLR, Phys. Rev. Letters, Phys. Rev. X, Phys. Rev. X Life, Phys. Rev. E, Entropy, Granular Matter
Identified inconsistency in publication leading to retraction (Phys. Rev. E (2020))
- **Editor** – Interactive explorable series “DataPoints”, Data Driven Discovery Institute, University of Pennsylvania (2024-2025)
 - *Created inaugural post, [Where is the information in data?](#)*
- **Organizer**, APS March Meeting focus session on “Information theory and physics” (2023, 2024)
- **Co-organizer**, APS March Meeting tutorial: “*Adventures with computational modeling and information theory*” (2023) [[website](#)]

Outreach and community service

- **Co-creator and Instructor** – Bicycle Physics summer program, Blackstone Bicycle Works, Chicago (2015-2017)
- **Instructor** – REU workshop on machine learning, UPenn (2023-2025)
- **Instructor** – Data Hangouts for undergraduates, UPenn (2022-2025)
- **Instructor** – Sustainability Training, Google NYC (2021)
- **Instructor** – “Paradigm shift” middle school program, UChicago (2014)
- **Co-organizer** – Soft matter physics journal club, UChicago (2017-2019)
- **Graduate Mentor** – UChicago Bridge Program for URM PhD students (2016)
- **Onboarding Mentor** – AI Residency, Google NYC (2020-21)
- **Undergraduate Mentor** – Supervised 12 students on 10+ projects, UChicago (2014-2019)